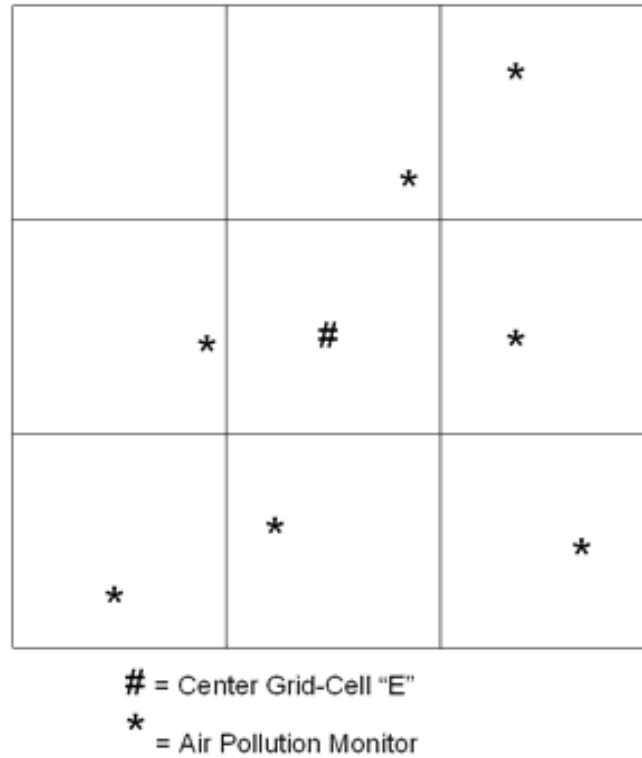
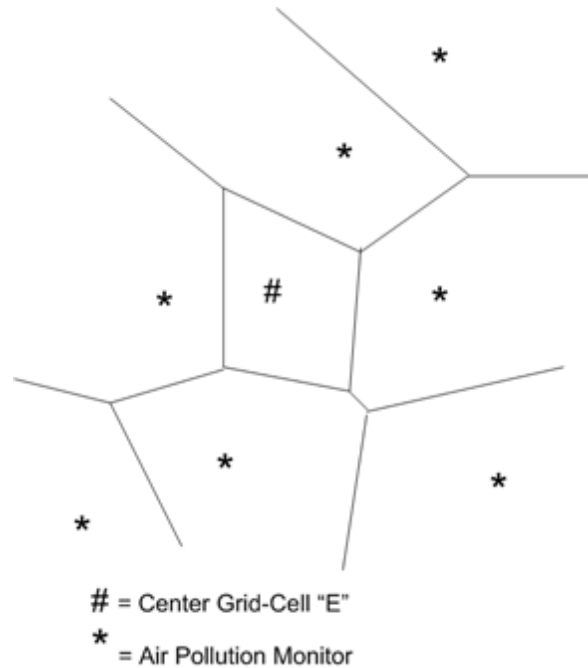
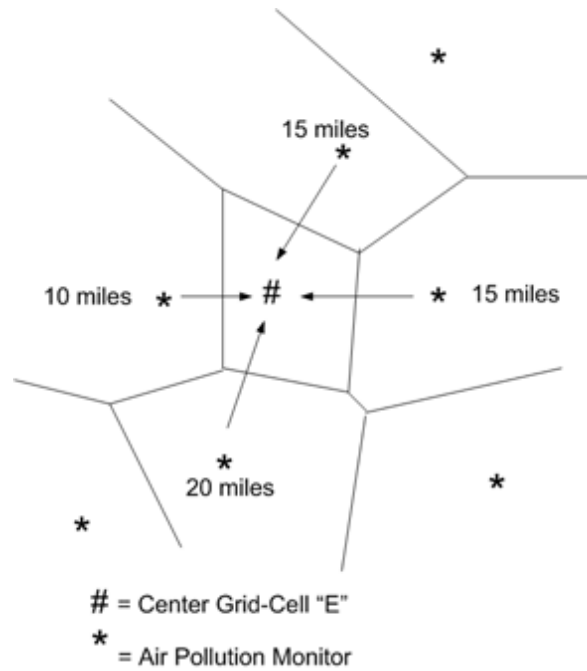




In particular, BenMAP identifies the nearest monitors, or “neighbors,” by drawing a polygon, or “Voronoi” cell, around the center of each BenMAP grid cell. The polygons have the special property that the boundaries are the same distance from the two closest points.



BenMAP then chooses those monitors that share a boundary with the center of grid-cell “E.” These are the nearest neighbors, BenMAP uses these monitors to estimate the air pollution level for this grid-cell.



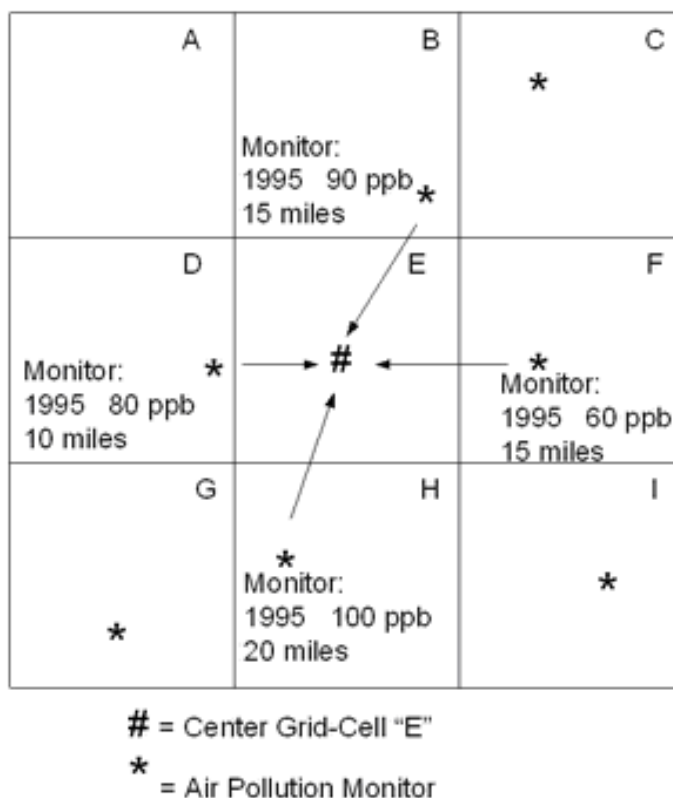
To estimate the air pollution level in each grid-cell, BenMAP calculates the metrics for each of the neighboring monitors, and then calculates an inverse-distance weighted average of the metrics. The further the monitor is from the BenMAP grid-cell, the smaller the weight.

In the figure below, the weight for the monitor 10 miles from the center of grid-cell E is calculated as follows:

$$weight_{10miles} = \frac{\frac{1}{10}}{\left(\frac{1}{10} + \frac{1}{15} + \frac{1}{15} + \frac{1}{20}\right)} = 0.35$$

The weights for the other monitors would be calculated in a similar fashion. BenMAP would then calculate an inverse-distance weighted average for 1995 air pollution levels in grid-cell E as follows:

Forecast 1995 = $0.35 \times 80 \text{ ppb} + 0.24 \times 90 \text{ ppb} + 0.24 \times 60 \text{ ppb} + 0.18 \times 100 \text{ ppb} = 81.2 \text{ ppb}$.



B.4 Fixed Radius

When using the fixed radius option to represent air pollution levels at a population grid-cell, BenMAP identifies all monitors within a specified distance of the center of the population grid cell, calculates the metrics at each monitor, and then calculates a weighted average of the metrics using the algorithms described for VNA. When no monitors are within the specified distance, BenMAP assigns the closest monitor to a population grid-cell, and calculates the metrics using the algorithms described for the closest monitor approach.

B.5 Monitor or Model Data with Missing Days

When estimating air pollution exposure, it will often happen that metrics are missing for one or more days in the season or year. To remedy this, BenMAP calculates seasonal average values to substitute for missing daily values within each season. When combining air pollution metrics from multiple monitors, BenMAP first calculates the seasonal average values for the daily metrics, substitutes these for missing values, and then performs the user-specified interpolation method, such as VNA.

Appendix C. Deriving Health Impact Functions

This Appendix provides of an overview regarding the health impact functions that BenMAP uses to estimate the impact of a change in air pollution on adverse health effects. It provides a description of the particular types of health impact functions that BenMAP uses.

The functional form of the relationship between the change in pollutant concentration, Δx , and the change in population health response (usually an incidence rate), Δy depends on the functional form of the C-R function from which it is derived, and this depends on the underlying relationship assumed in the epidemiological study chosen to estimate a given effect. For expository simplicity, the following subsections refer simply to a generic adverse health effect, “y” and uses particulate matter (PM) as the pollutant - that is, $\Delta x = \Delta \text{PM}$ - to illustrate how the relationship between Δx and Δy is derived from each of several different C-R functions.

Estimating the relationship between ΔPM and Δy can be thought of as consisting of three steps:

- (1) choosing a functional form of the relationship between PM and y (the C-R function),
- (2) estimating the values of the parameters in the C-R function assumed, and
- (3) deriving the relationship between ΔPM and Δy (the health impact function) from the relationship between PM and y (the C-R function).

Epidemiological studies have used a variety of functional forms for C-R functions. Some studies have assumed that the relationship between adverse health and pollution is best described by a linear form, where the relationship between y and PM is estimated by a linear regression in which y is the dependent variable and PM is one of several independent variables. Log-linear regression and logistic regression are other common forms.

Note that the log-linear form used in the epidemiological literature is often referred to as “Poisson regression” because the underlying dependent variable is a count (e.g., number of deaths), believed to be Poisson distributed. The model parameters may be estimated by regression techniques but are often estimated by maximum likelihood techniques. The form of the model, however, is still log-linear.

C.1 Overview

The relationship between the concentration of a pollutant, x, and the population response, y, is called the concentration-response (C-R) function. For example, the concentration of fine particulate matter (PM_{2.5}) may be in $\mu\text{g}/\text{m}^3$ per day, and the population response may be the number of premature deaths per 100,000 population per day. C-R functions are estimated in epidemiological studies. A functional form is chosen by the researcher, and the parameters of the function are estimated using data